

Jack Nystrom

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Education

University of California Santa Cruz

Anticipated Graduation: June 2027

B.S. in Computer Science | Electrical Engineering Minor

GPA: 4.0

Coursework: Computer Architecture, C Programming, Computer Systems and Assembly, Data Structures and Algorithms

Experience

FSAE Electric Team, Formula Slug | *Firmware Lead*

Santa Cruz, CA | **September 2024 – Present**

- Lead firmware development, including battery management, throttle control, data logging, and CAN bus design
- Organize PCB testing, sensor integration, wire harnessing, telemetry system development, and data analysis
- Develop traction control algorithm utilizing wheel speed, IMU, and inverter data to calculate active slip prevention
- Teach embedded C++ practices, achieving 100% static memory, and enforcing non-blocking I/O and proper ISRs
- Caught multiple isoSPI design mistakes with 2026 BMS board design, enabling timely revision & battery completion
- Transitioned team to latest MbedOS RTOS, developing custom board targets with clock configs and CMake patches
- Ported new IMU's SDK to MbedOS RTOS, implementing a C++ shim layer for its threading and UART APIs
- Fixed throttle control bugs at 2025 competition, enabling us to pass inspection and compete for the first time ever

Zero Motorcycles | *Software Intern*

Scotts Valley, CA | **June 2025 – Present**

- Develop customer-facing motorcycle debugging software, enabling successful 1.0 release to bike dealers
- Improve reliability and implement new features in all areas, from CAN communication and ECU firmware updates to cloud integration and Flutter UI development
- Develop CI/CD infrastructure and internal tooling to streamline testing and development of the diagnostic software

Electronics Workshop | *Course Development Volunteer*

Albany, CA | **August 2023 – June 2024**

- Assisted creating new Electronics Workshop high school course, leading curriculum and project design
- Developed MicroPython libraries for the motor controller, IMU, distance and color sensors used in class projects
- Authored 3,000-word course documentation detailing all hardware and software used

Projects

Telemetry Peripheral PCB (Formula Slug)

November 2024 – June 2025

- Developed peripheral data acquisition PCB in KiCad, with an AVR microcontroller and external CAN controller
- Created a driver in C from scratch for the MCP2515 SPI-based CAN controller via AVR SDK, w/ clock prescaling
- Implemented strain gauge amplifier, frequency-voltage converter to process & log sensor data in small form factor

STM32 Board Template (Formula Slug)

June 2025 – September 2025

- Created STM32 design template and PCB with KiCad, enabling transition away from Nucleo development boards
- Developed custom MbedOS board targets for all racecar PCBs, enabling pure-CMake build system for firmware
- Implemented switching DC-DC converter, LDO, CAN transceiver, and high-spec and affordable STM32
- Significantly reduced board size and cost from past years while maintaining board design simplicity

Data Visualization Stack (Formula Slug) ([link](#))

September 2024 – Present

- Developed telemetry and data visualization systems, to provide instant and accessible insights for team members
- Created full-stack pipeline from C++ data logging firmware to high-performance browser visualizations in React
- Designed custom columnar binary data logging format, enabling easy on-car decoding of CAN into data tables

Portfolio Website (bvnggee.com)

June 2023 – Present

- Teach readers about software engineering through technical educational blog posts on a variety of unique subjects
- Reduced site deployment time from 3 minutes to 5 seconds using a custom pipeline with Nix, Nginx, Bash & Docker

Minecraft Mods ([link](#))

August 2020 – June 2022

- Added rendering improvements, gameplay features, accessibility & UI enhancements, and bug fixes to Minecraft
- Gained over 16 thousand downloads and hundreds of positive reviews on CurseForge

Skills

Areas: Software Engineering, Embedded Programming, PCB Testing, Digital/Analog Circuit Design

Programming Languages: C, C++, CMake, Python, Rust, Lua, Java, Bash, JS/TS

Digital Protocols: CAN, I2C, SPI, UART, USB, 1Wire

Tools: Oscilloscope, Multimeter, Waveform Generator, Logic Analyzer, Power Supplies, ADC&DAC, THT&SMT Soldering

Software: KiCad, Linux, Git, GDB, Docker, CLIs, Nix